

PORTFOLIO EXECUTIVE DECK

International Basketball Analytics

20 Years of FIBA Competitions (2005-2024): An End-to-End Data, ML & Decision Support System

Python + R + DuckDB + Parquet | 1,145 Games | 27,353 Player Records | 227 Automated Tests

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Open-Source Repository: <https://github.com/miguelo0203/basketball-analytics>

Executive Summary: From Raw Data to Court Decision

A complete data engineering and predictive pipeline built for professional basketball decision-makers.

01. Data Foundation

18 senior FIBA tournaments (EuroBasket, World Cup, Olympics) structured into an OLAP DuckDB warehouse with 12 normalized tables.

02. Advanced Metrics

Dean Oliver Four Factors normalized to pace, True Shooting %, and longitudinal career stability via Bayesian shrinkage.

03. Machine Learning

Chronological 17-fold walk-forward validation (1,105 test games). Brier Score 0.1967 (vs 0.2500 naive) and ECE 0.0314.

04. Decision Support

1.5-page pre-game coaching briefs, 6 objective player archetypes, and anti-hindsight validation mode.

The Analytics Challenge in Professional Basketball

Why traditional boxscores and 40-page scouting dossiers fail modern coaching staffs.

The Status Quo: Information Overload

- Scouting reports are flooded with raw totals (PPG, RPG) that ignore pace and lineup context.
- Short-tournament variance (6-9 games) misleads decision-makers into over-valuing shooting hot streaks.
- Models trained with random k-fold suffer from temporal data leakage, learning future tactical eras.
- Analysts deliver disconnected charts instead of structured pre-game decision questions.

Our Solution: The Data-First Pipeline

- Pace-neutral Four Factors isolate true possession efficiency.
- Bayesian shrinkage ($\lambda = 0.75$) stabilizes small-sample noise toward historical priors.
- Strict walk-forward expanding window guarantees zero future contamination.
- Concise 1.5-page coaching briefs delivering actionable tactical hypotheses before tip-off.

01

Data Architecture & Engineering

From Raw Historical Archives to In-Process Columnar OLAP with DuckDB and Parquet

Historical Scope & Canonical Scale

Two decades of international men's senior basketball verified with SHA-256 cryptographic provenance.

1,145

OFFICIAL MATCHES

Every game across 18 FIBA tournaments (2005-2024).

18

TOURNAMENTS

EuroBasket (2005-2022), FIBA World Cup (2006-2023), Olympics (2008-2024).

27,353

PLAYER PERFORMANCES

Individual boxscore game logs across all 18 tournaments.

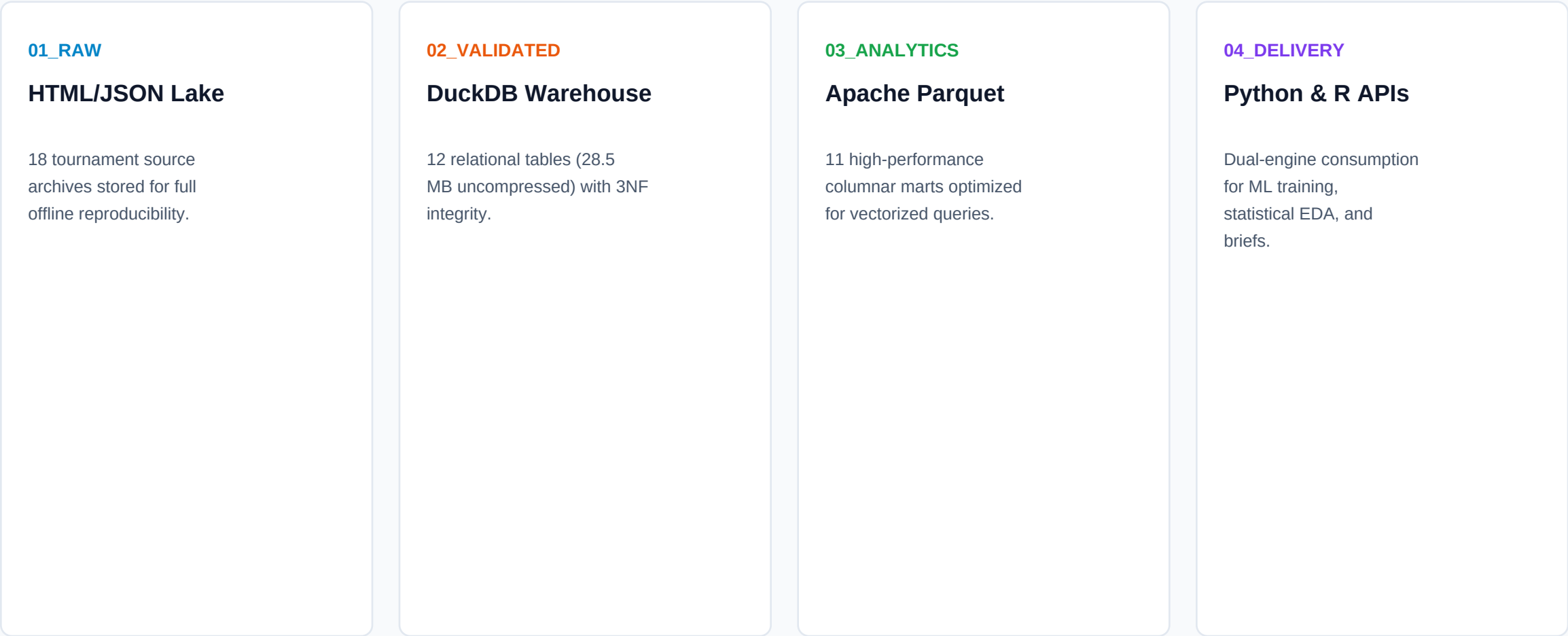
2,124

CANONICAL PLAYERS

Deterministic entity resolution mapping names across 20 years.

Medallion Data Warehouse Architecture

Decoupled pipeline separating ingestion, relational validation, columnar storage, and analytics.



Performance: Sub-millisecond aggregations on 27k+ player rows directly in-memory.

Automated Quality Assurance & Invariants

Rigorous mathematical contracts enforced prior to any statistical modeling.

200 Minutes / Game Rule

Every 40-minute regulation game strictly verifies exactly 200 player-minutes (plus 25 min per 5-min overtime). Any mismatch fails the pipeline.

Score & Boxscore Closure

Sum of points ($2P + 3P + FT$) is reconciled against official final match scores. Zero scoring discrepancies across 1,145 games.

Entity Resolution Engine

Unified 2,124 players across language variants, accents, and name changes (e.g. Dirk Nowitzki, Bojan vs Bogdan Bogdanovic) using custom slug hashing.

02

Analytics & Modeling Engine

Pace Normalization, Longitudinal Stability, Player Archetypes, and Calibrated Machine Learning

Pace Normalization & Dean Oliver Four Factors

Decomposing basketball efficiency into four fundamental, pace-neutral drivers.

Effective FG% (eFG%) (Shooting Efficiency)

Weights 3-pointers at 1.5x. Accounts for shot quality differences.

$$eFG\% = (FGM + 0.5 * 3PM) / FGA$$

Turnover Rate (TOV%) (Ball Security)

Measures possessions ending in turnover per 100 plays.

$$TOV\% = TOV / (FGA + 0.44 * FTA + TOV)$$

Off. Rebounding (ORB%) (Second Chances)

Percentage of available offensive rebounds secured.

$$ORB\% = ORB / (ORB + Opp_DRB)$$

Free Throw Rate (FTR) (Foul Drawing)

Ratio of free throw attempts to field goal attempts.

$$FTR = FTA / FGA$$

Overcoming Small-Sample Tournament Noise

Bayesian shrinkage ($\lambda = 0.75$) stabilizes noisy tournament shooting toward career priors.

ESTIMATION FLOW: FROM NOISE TO STABILITY

*** 1. RAW TOURNAMENT SIGNAL**

7-game sample: 15/30 3PT = 50.0% (Small-Sample Trap)

v 2. HIGH VARIANCE & NOISE

A swing of just 3 shots drops 3PT% from 50% to 40%

v 3. BAYESIAN SHRINKAGE ($\lambda = 0.75$)

Weighted contraction toward historical career baseline

v 4. STABILIZED ESTIMATE

Reliable true skill projection for pre-game tactical planning

$\lambda = 0.75$ | BAYESIAN SHRINKAGE FACTOR

Bayesian shrinkage weight contracting noisy short-sample shooting estimates toward 3,767 player career baselines.

[OK] High-volume samples shrink less; short streaks regress.

CONCEPTUAL SHRINKAGE SPECTRUM

-> High Variance / Noise

-> Reliable Game-Plan Prior

-> Historical Baseline

Cluster Bootstrap (B = 5,000):

Non-parametric 95% confidence intervals across 3,767 qualified campaigns.

6 Objective Functional Archetypes

Clustering 3,767 player campaigns via K-Means++ and PCA on 14 normalized per-40 metrics.

Primary Initiator

High usage, primary P&R handler, elite shot creation and passing volume.

Floor Spacer

High 3P volume, catch-and-shoot specialist, low interior touches, elite spacing.

Interior Hub

Post playmaker, rim finisher, high offensive rebounding and short-roll passer.

Floor General

Pure playmaker, exceptional assist-to-turnover ratio, low turnover rate, tempo controller.

Defensive Anchor

Rim protector, elite defensive rebounder, screen-and-roll finisher, low usage.

Balanced Wing

Versatile two-way wing, secondary creator, switchable defender, balanced shot profile.

Calibrated Machine Learning: LightGBM

Probabilistic match forecasting regularized with L2 penalty and isotonic calibration.

0.1967

BRIER SCORE

vs 0.2500 naive baseline (+21.3% quadratic error reduction).

0.0314

CALIBRATION ERROR (ECE)

Expected Calibration Error: within 3.14% of perfect calibration.

11.74

POINT MARGIN MAE

Mean Absolute Error in point spread across 1,105 test matches.

17-Fold Temporal Walk-Forward Protocol

Strict expanding-window splits preventing temporal data leakage.

Fold 01

Train: EuroBasket 2005

-> Test: FIBA World Cup 2006 (Out-of-sample)

Fold 05

Train: 2005-2009 Tournaments

-> Test: FIBA World Cup 2010 (Out-of-sample)

Fold 11

Train: 2005-2016 Tournaments

-> Test: FIBA EuroBasket 2017 (Out-of-sample)

Fold 17

Train: 2005-2023 Tournaments

-> Test: Paris Olympic Games 2024 (Out-of-sample)

Monte Carlo Tournament Simulation Engine

Simulating full tournament outcome distributions across 180,000 bracket iterations.

SIMULATION PIPELINE: Historical Data -> Strength Priors -> Monte Carlo (180,000) -> Probabilistic Outcomes

DYNAMIC TOURNAMENT BRACKET FLOW

> Round of 16

Match 01 | Match 02 | Match 03 | Match 04

> Quarterfinals

QF 01 | QF 02

> Semifinals

SF 01 | SF 02

> Final & Podium

CHAMPION | MEDAL TREE

180,000 | BRACKET ITERATIONS

Full dynamic bracket simulations accounting for match sequence and cumulative fatigue.

lambda = 0.75 | BAYESIAN SHRINKAGE

Prevents over-reacting to preliminary round group-stage upsets.

PROBABILISTIC | OUTCOME DISTRIBUTIONS

Outputs conditional podium probability trees rather than fragile single-winner picks.

Interactive Workspace & Anti-Hindsight Protocol

Isolating pre-game evidence from post-game outcomes to eliminate hindsight bias.

[APP] ANALYST DECISION WORKSPACE | Streamlit Decision Support Engine

APP FILTERS

- Target Match: Spain vs USA
- Checkpoint: T-1 (Pre-Game)
- Evidence: Four Factors & Drop
- Roster State: Canonical 2008
- Status: ANTI-HINDSIGHT ACTIVE

PRE-GAME EVIDENCE (AVAILABLE)

- Pace-neutral Four Factors (+4.2 Net Rating half-court)
- Opponent defense alert: Center deep drop on P&R
- Longitudinal shooting priors ($\lambda = 0.75$ stabilized)
- Functional archetype matchup synergy & rim delta

POST-GAME (QUARANTINED)

[QUARANTINED POST-GAME INFORMATION]

- Final score & official boxscore totals
- Shooting percentages of current game
- Post-hoc tactical rationalizations

ANTI-HINDSIGHT AUDIT TRAIL: T-7 Pre-Brief -> T-1 Decision Log -> Game Tip-Off -> Outcome Revealed -> Post-Game Audit

03

Selected Case Studies

Four Applied Demonstrations: Tactical Support, Data Engineering, ML Rigor, and Longitudinal R Analysis

From Data to the Whiteboard: Beijing 2008 Final

How pre-game analytics framed the tactical strategy for Spain vs. USA 'Redeem Team'.

Quantitative Pre-Game Signals

- Context: Spain lost by 37 pts (82-119) in group phase against USA.
- Half-Court Signal: In 5v5 half-court sets, Spain held a +4.2 Net Rating advantage via Gasol brothers inside.
- Transition Vulnerability: USA scored 1.25 PPP on transition off turnovers.
- Defensive Flaw: USA bigs dropped deep into paint to pack driving lanes.

Tactical Whiteboard Brief

- Pace Control: Limit total possessions below 72 using 16+ second attacks.
- Transition Defense: Implement 2-3 matchup zone immediately after made baskets.
- Pick-and-Pop Spacing: Exploit deep drop with 3-pointers from Pau, Marc, and Garbajosa.
- Court Outcome: Spain cut deficit to 4 pts (108-104) at 2:20; final 107-118.

In-Process OLAP Lakehouse with DuckDB & Parquet

Deterministic data engineering for 20 years of heterogeneous international tournaments.

Engineering Challenges

- Heterogeneous FIBA Boxscores across 2005-2024 with varying column conventions.
- Diacritic Name Variations across international alphabets (Nowitzki, Spanoulis, Bogdanovic).
- Zero Cloud Infrastructure cost constraint: 100% offline local reproducibility.
- Dual-Language Access: Zero-copy querying from both Python and R.

Architecture Solutions

- DuckDB Relational Engine: 12 tables, Star Schema, in-process ACID transactions.
- 11 Apache Parquet Marts: Snappy-compressed columnar storage with SHA-256 hashes.
- Deterministic Entity Resolver: 2,124 canonical player IDs without paid commercial APIs.
- Automated QA Suite: 227 tests in pytest enforcing 200 min/game invariants.

Zero-Leakage ML with 17 Chronological Folds

Avoiding temporal leakage and delivering calibrated probabilities for tournament play.

Methodological Traps Avoided

- Random K-Fold Cross-Validation leaks future playstyle evolution into past evaluations.
- Over-confident predictions: raw neural nets produce uncalibrated probabilities in short events.
- Overfitting on small samples: complex models over-index on single-game tournament anomalies.
- Causal Overclaiming: confusing feature importance with tactical ground truth.

Rigorous ML Protocol

- Expanding Window: 17 folds evaluating strictly unseen tournaments (1,105 test games).
- LightGBM with L2 Regularization on pre-game Four Factors differentials.
- Isotonic Probability Calibration: Brier Score 0.1967 (vs 0.2500 naive), ECE 0.0314.
- TreeSHAP Attribution: Interpretable local feature contribution without causal exaggeration.

Longitudinal Shooting & Role Mining in R / Quarto

Multi-year player evaluation, non-parametric bootstrap, and reproducible reporting.

Statistical Analysis in R

- Direct DuckDB Connection via DBI: Zero data duplication between Python and R.
- Longitudinal Career Tracking: 3,767 qualified player campaigns (≥ 40 minutes).
- Cluster Bootstrap ($B = 5,000$): Non-parametric 95% confidence intervals for True Shooting %.
- Rebounding & Turnover Stability: Higher year-over-year correlation ($r > 0.65$) than 3P%.

Clustering & Quarto Reporting

- PCA Decomposition: First 3 principal components explain $>60\%$ of player variance.
- K-Means++ Clustering: Discovery of 6 functional archetypes based on 14 normalized metrics.
- Quarto CLI Pipeline: Automated compilation into interactive HTML report with ggplot2.
- Custom Visualization: ``theme_basketball_analytics()`` with 300 DPI vector exports.

04

Validation & Engineering Rigor

227 Automated Pytest Tests, Cross-Language Verification, and Deterministic Master Runner

227 Automated Regression Tests (100% Pass)

Continuous integration test suite covering 26 specialized test modules.

Data Integrity & Invariants

Validates 200 min/game, boxscore closure, SHA-256 hashes, DuckDB schemas, and column contracts across all 18 tournaments.

Analytical & ML Precision

Tests walk-forward temporal splits, Brier Score bounds, ECE limits, Monte Carlo distribution bounds, and SHAP attribution coherence.

Portfolio & Cross-Language

Verifies R-DuckDB parity, Quarto compilation, publication package completeness, and zero prohibited marketing buzzwords.

Cross-Language Parity: Python + R + DuckDB

Ensuring identical analytical results regardless of execution environment.

Unified Warehouse	Single <code>`basketball_analytics.duckdb`</code> file accessed concurrently by Python (<code>`duckdb`</code>) and R (<code>`DBI::dbConnect`</code>).
Zero Discrepancy	Exact match on 1,145 games, 2,290 team rows, 27,353 player boxscores, and 4,350 qualified records.
Metric Synchronization	Identical eFG% (0.5355), Pace (61.72), and 3P Attempt Rate (0.3153) across Python and R pipelines.
Decoupled Design	Python handles ETL/ML; R handles statistical inference and visualization; Parquet provides columnar handoff.

Deterministic End-to-End Reproducibility

Run the entire project verification pipeline with a single master command.

```
$ python scripts/run_project.py
```

1. Environment Check

Verifies Python 3.10+, R 4.4+, DuckDB, PyArrow, LightGBM, Pandas, and Pytest.

2. Warehouse Check

Validates DuckDB database (16.8 MB) and 11 Parquet analytical marts.

3. R Analysis Pipeline

Executes R statistical scripts and generates figures in reports/figures_r/ in ~21s.

4. Pytest Regression Suite

Runs all 227 automated tests with 100% pass rate in ~2 minutes.

05

Limitations & Professional Scope

Scientific Honesty, Transparent Boundaries, and Operational Value for Basketball Clubs

Transparent Boundaries: What the System Claims

Clear distinction between data-backed capabilities and operational limits.

What the System DEMONSTRATES

- Pace-neutral Four Factors and True Shooting evaluation.
- Longitudinal Bayesian shrinkage to control short-tournament variance.
- Strict out-of-sample calibrated match probability modeling.
- Structured pre-game briefs that ask the right tactical questions.
- Fully reproducible in-process OLAP data lakehouse.

What the System DOES NOT Claim

- No 25Hz live optical tracking (no Second Spectrum camera telemetry).
- No automatic game decisions: data structures evidence, coach decides.
- No infallible betting prediction: models estimate uncertainty.
- No causal guarantees: statistical associations are not mechanical laws.
- No real-time in-game bench tracking (focused on pre/post game).

Operational Value by Basketball Role

How different stakeholders in a basketball club extract value from this platform.

ORGANIZATIONAL IMPACT: BASKETBALL CLUB DECISION ECOSYSTEM

HEAD COACH & STAFF

"WHAT TO CHANGE"

- Pre-game 1.5-page coaching briefs
- Opponent P&R drop coverage alerts
- Pace & possession targets (<= 72 poss)

SCOUTING DEPARTMENT

"WHO FITS"

- 6 objective functional archetypes
- Small-sample shooting shrinkage
- Roster complement & role audit

SPORTING DIRECTOR / GM

"WHAT TO INVEST IN"

- Evidence-based contract valuation
- Career trajectory stability curves
- Recruitment risk minimization

VALUE PIPELINE: RAW DATA (DuckDB Lakehouse) -> EVIDENCE (R & Calibrated ML) -> DECISION SUPPORT (Pizarra & Roster)

Day-1 Club Onboarding Roadmap (30 Days)

How this modular platform deploys seamlessly in domestic club leagues.

TRANSITION TIMELINE: DAY 01 -----> DAY 10 -----> DAY 20 -----> DAY 30

PHASE 01 : DATA AUDIT

Days 1-10

- Connect domestic boxscores (Synergy, Genius, EuroLeague API)
- Validate relational DuckDB schema & table contracts
- Execute automated QA suite (200 min/game invariants)

PHASE 02 : ANALYST WORKFLOW

Days 11-20

- Calibrate league-specific Four Factors baselines
- Set Bayesian shrinkage priors ($\lambda = 0.75$) for roster
- Train video analysts on anti-hindsight workspace

PHASE 03 : STAFF INTEGRATION

Days 21-30

- Deliver automated pre-game briefs directly to coaching staff
- Integrate Four Factors alerts into whiteboard pre-game talks
- Establish weekly post-game accuracy review loop

[*] DAY 30 MILESTONE: FULLY EMBEDDED DECISION SYSTEM (Zero Cloud Overhead, 100% Offline Reproducibility)

Summary & Key Takeaways

A disciplined, reproducible approach to basketball quantitative analytics.

1. Data-First & Rigorous

20 years of FIBA tournaments structured in DuckDB and Parquet with 227 automated tests and zero data leakage.

2. Calibrated Machine Learning

Brier Score 0.1967 and ECE 0.0314 across 17 chronological folds (1,105 test matches).

3. Actionable Tactical Impact

1.5-page briefs that ask concrete questions for the whiteboard instead of dumping raw statistical charts.

4. Transparent & Reproducible

Complete open-source pipeline executed with a single command on GitHub.

PROJECT REPOSITORY & CITATION

International Basketball Analytics

Explore the Full Open-Source Codebase, Case Studies, and Automated Tests

GitHub Repository: <https://github.com/miguelo0203/basketball-analytics>

Release v1.0.0 (Data-First, Verified, MIT License)

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